

MILESTONE 03 · AI/ML ARCHITECTURE

ICAP

Intelligent Cognitive Alarm Platform

Cognitive Challenge Engine & Dynamic Difficulty Adaptation

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7

Cognitive Domains

5

Dynamic Elo Tiers

92.5%

Snooze AI Accuracy

100%

Verification Rate

Milestone 3 — AI/ML Transformation

From static models to live operational microservices

CORE OBJECTIVE

In Milestone 3, the AI/ML subsystem transitioned into production-ready microservices. We delivered the Cognitive Challenge Engine with dynamic Elo difficulty scaling, multi-answer set validation, and server-side anti-cheat protection.

5

Core AI Modules

Fully Operational & Deployed

MAJOR MILESTONE DELIVERABLES

Cognitive Challenge Engine

7 procedurally generated categories across 5 dynamic Elo difficulty tiers

Live FastAPI Microservice Serving

GET /users/{id}/challenge & POST /challenge/validate connected to frontend

Adaptive Elo Rating Engine

Continuous reinforcement learning recalibration per challenge attempt

Server-Side Anti-Cheat Security

Zero-leak response payload & single-use challenge token popping

Multi-Answer Validation Engine

Set-based synonym & antonym resolution for natural linguistic tolerance

My Core Engineering Contributions

AI/ML Developer — Algorithms, security, and live platform integration

AI/ML Algorithmic Engineering

- ◆ Engineered 7 procedural Cognitive Challenge generators with zero static repetition
- ◆ Implemented Reinforcement Learning Elo engine spanning 5 dynamic difficulty tiers
- ◆ Built multi-answer set validation engine resolving synonym and antonym ambiguities
- ◆ Trained & serialized production model weights (snooze_predictor, difficulty_model)
- ◆ Engineered behavioral feature extraction pipeline in features.py for habit scoring

Platform Serving & Microservices

- ◆ Exposed 3 live REST HTTP endpoints via FastAPI for frontend mobile consumption
- ◆ Integrated directly with React Native mobile client for challenge selection & dispatch
- ◆ Implemented server-side answer suppression & single-use challenge token popping
- ◆ Connected PostgreSQL & MongoDB databases via unified data_ingest.py pipeline
- ◆ Achieved 100% test pass rate across all verification suites in FastAPI TestClient

Cognitive Challenge Generation Engine

Procedural multi-domain intelligence for anti-habituating wakefulness

- ◆ Procedurally generates unique challenges across 7 cognitive domains at runtime using dynamic seeds
- ◆ Eliminates hardcoded repetition — prevents users from answering on autopilot half-asleep
- ◆ Consumes real-time user category preferences from MongoDB user_preferences
- ◆ Dispatches dynamic challenge payloads directly to the React Native lock-screen UI

7 PROCEDURAL COGNITIVE DOMAINS

- ◆ Mathematical Problems (Arithmetic, Percentages, Algebra)
- ◆ Logic Puzzles (Sequences, Missing Element Series)
- ◆ Memory Challenges (Sequential Word List Recall)
- ◆ Word Games (Anagrams, Synonyms, Antonyms)
- ◆ Pattern Recognition (Shape & Geometric Progression)
- ◆ Lateral Riddles (Creative Logic Thinking)
- ◆ Quick Quizzes (General Science & Fact Trivia)
- ◆ Zero Repetition Guarantee (Algorithmic Generation)

Adaptive Difficulty & Elo Scaling Engine

Continuous skill tracking & dynamic bracket recalibration

- ◆ Reinforcement learning Elo engine recalculates user rating after every single challenge attempt
- ◆ Target solve window: 15–30 seconds — fast solves elevate Elo rating, slow/wrong solves decrease Elo
- ◆ Dynamically maps continuous Elo ratings into 5 discrete baseline difficulty tiers in real time
- ◆ Maintains optimal cognitive challenge level: prevents boredom for experts and morning fatigue for beginners

5 DYNAMIC DIFFICULTY TIERS MAPPING

BEGINNER

< 900 Elo

Single-digit arithmetic,
basic antonyms

EASY

900–1100 Elo

Two-step operations,
simple word puzzles

MEDIUM

1100–1350 Elo

Geometric series,
5-word memory recall

HARD

1350–1600 Elo

Algebraic equations,
complex riddles

EXPERT

> 1600 Elo

Advanced lateral logic,
multi-step patterns

REST API Serving & Anti-Cheat Security

Enterprise-grade endpoint protection and live mobile consumption

Server-Side Anti-Cheat Protocols

- ◆ Zero-Leak Response: 'correct_answer' omitted from GET payload
- ◆ One-Time Token Popping: Challenge token deleted on validate attempt
- ◆ Multi-Answer Set: accepted_answers handles synonyms (e.g., 'heavy')
- ◆ Prevents client-side network inspection, answer sniffing, and replay attacks
- ◆ Enforces strict single-attempt challenge verification lifecycle

Live REST Endpoints (FastAPI)

- ◆ GET /users/{user_id}/challenge → Dispatches personalized challenge by Elo
- ◆ GET /challenge → Dispatches generic challenge with query overrides
- ◆ POST /challenge/validate → Securely verifies submitted response
- ◆ Seamlessly integrated with mobile React Native Challenge Selection screen
- ◆ Verified live with 200 OK & 100% test pass rate in Swagger UI console

AI/ML Subsystem Architecture & Tools

Production-grade scientific computing and async web services

Python 3.13

Core computational engine

FastAPI & Uvicorn

High-performance async REST microservices

Pydantic v2

Strict schema validation & serialization

XGBoost Classifier

Snooze oversleeping risk model (92.5%)

Scikit-learn

ML preprocessing & Elo modeling algorithms

Pandas & NumPy

Feature engineering & matrix vectorization

Joblib Persistence

Binary serialization of trained model weights

Swagger UI / OpenAPI

Interactive API contract & endpoint testing

Future Work & AI/ML Evolution

Transition plan towards production deployment and multi-modal models

01

Transition in-memory challenge caching to persistent MongoDB challenge_data collection

02

Perform full end-to-end integration and load testing with backend gateway and mobile client

03

Fine-tune XGBoost snooze prediction hyperparameters with live historical user telemetry

04

Implement multi-lingual challenge procedural generation across major Indian regional languages

05

Deploy complete ICAP AI/ML microservice container stack for final production evaluation

Conclusion & Milestone 3 Delivery

AI/ML subsystem is 100% verified, secure, and ready for production



Successfully implemented and delivered the Cognitive Challenge Engine for ICAP Milestone 3



Engineered 7 procedurally generated challenge categories across 5 dynamic Elo difficulty tiers



Exposed and validated 3 live REST HTTP endpoints integrated directly with the mobile client



Built robust server-side answer protection, one-time token popping, and multi-answer validation



Achieved 100% test pass rate across all verification suites, establishing a reliable AI foundation